# ASSIGNMENT NO 02

# (Project Report)

**Group Name:**

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**Reg No :**

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**Section:**

**SE-3B**

**Course:**

**Introduction to Software Engineering**

## Project Overview:

This project is a **Loan Calculator application** that can compute various loan-related values, including monthly payments, remaining balance, number of payments, interest rates, and total loan amount. It offers both a **Graphical User Interface (GUI)** and a **Command Line Interface (CLI)**.
In the **GUI version**, users can enter their loan details and instantly view the results in a visual format. The **command-line mode** allows users to perform calculations by passing different options and arguments.

The main components of the project include:

- A central class, **LoanCalculator**, responsible for all financial computations.
- A **Qt-based GUI application** (`LoanCalcQtMainWindow`) for interactive use.
- A **command-line interface** that performs calculations based on user-specified options.
- The ability to **load loan details from a configuration file** (`config.txt`).
- **Unit tests using GoogleTest** to verify correctness and ensure stability.
- **Doxygen documentation** to generate clean and organized HTML documentation automatically.

The goals of this project are to:

- Provide accurate and reliable loan calculations.
- Offer flexible ways for users to run the tool, either through GUI or CLI.
- Test key scenarios, such as normal EMI calculations, handling invalid inputs, and working with large loan tenures.
- Maintain clean, well-documented, and easily understandable code for future updates and improvements.

## Code snippets of bug fixes and refactorings:

## Bug 1: EMI calculation returns NaN or -NaN

**Problem:**

- `calculatePayment()` returned `nan` for normal inputs like 100,000 principal and 6% interest.
- Also failed for large periods (like 120 months) or when interest is zero.

- Caused test failures in GoogleTest (`NormalEMICalculation`, `LargeTenureNoOverflow`).

**Cause:**

- Monthly interest was not correctly calculated from yearly interest.
- Formula for EMI didn't handle `0% interest` case.
- `powl` exponent term caused precision issues when used with float/double.

**Fix:**

- Use `long double` for all core calculations.
- Convert yearly interest to monthly decimal safely.
- Handle 0% interest separately to avoid division by zero.

## Code:

```
long double LoanCalculator::calculatePayment()
{
   long double principal = amount_ - initialPayment_;
   principal += openingFee_ + (principal * openingPercent_/100.0L);

   long double monthlyRate = (interest_ > 0) ? (interest_/100.0L)/12.0L : 0.0L;

   if(monthlyRate == 0.0L)
      return principal / periodTotal_;

   long double powTerm = powl(1.0L + monthlyRate, periodTotal_);
   long double emi = (monthlyRate * principal * powTerm) / (powTerm - 1.0L);

   return emi;
}
```

## Bug 2: GoogleTest `std::isnan` / `std::isinf` compilation errors

**Problem:**

- When running `LargeTenureNoOverflow` test, compiler gave:

error: ‘isnan’ is not a member of ‘std’

error: ‘isinf’ is not a member of ‘std’

**Cause:**

- `<cmath>` was not included in the test file properly.
- Some compilers require explicit `std::` with `isnan` and `isinf`.

**Fix:**

- Added `#include <cmath>` in `testLoan.cpp`.
- Used `std::isnan()` and `std::isinf()` to check floating-point results.

#include <cmath>

ASSERT_FALSE(std::isnan(emi));

ASSERT_FALSE(std::isinf(emi));

## Bug 3: Missing validation for negative inputs

**Problem:**

- Previously, users could set negative values for:
    - `amount_`, `interest_`, `payment_`
- This could produce invalid EMI, loan balance, or `log()` errors.

**Fix:**

- Added a helper `validatePositive()` method and used it in all setters

```
void LoanCalculator::validatePositive(long double value, const std::string &msg)
{
   if(value < 0)
      throw std::invalid_argument(msg);
}
```

```
void LoanCalculator::setAmount(long double A) {
    validatePositive(A, "Amount cannot be negative");
    amount_ = A;
}

void LoanCalculator::setInterest(long double i) {
    validatePositive(i, "Interest cannot be negative");
    interest_ = i;
    interestPeriodic_ = (i/100.0L)/12.0L;
}
```

## Bug 4: Principal calculation ignored initial payment & opening fee

**Problem:**

- EMI formula didn't account for `initialPayment_`, `openingFee_`, or `openingPercent_`.
- Test results and EMI output were incorrect for real scenarios.

**Fix:**

- Principal now properly includes all adjustments:

```
long double principal = amount_ - initialPayment_;
principal += openingFee_ + (principal * openingPercent_/100.0L);
```

## Bug 5: Loan balance and number of payments used inconsistent `interestPeriodic_`

**Problem:**

- `calculateLoanBalance()` and `calculateNumberPayments()` didn't safely cast interest to `long double`, causing rounding or NaN issues.

**Fix:**

- Converted all core formulas to use `long double`.

- Added domain checks to prevent log of non-positive numbers.

long double i = (long double)interestPeriodic_;

long double arg = 1.0L - (i * amount_ / payment_);

if(!(arg > 0.0L)) throw std::invalid_argument("Payment too small to repay loan");

## Refactorings:-

```
#ifndef LOANCALCULATOR_H

#define LOANCALCULATOR_H

#include <string>
#include <stdexcept>

class LoanCalculator
{
public:
// ---- Setters with validation (Bug 1 fix) ----
inline void setAmount(double A)
{
if (A <= 0)
throw std::invalid_argument("Loan amount must be > 0");
amount_ = A;
amountSet_ = true;
}

inline double getAmount() const { return amount_; }

// Initial down payment
inline void setInitialPayment(double initialA)
{
if(initialA < 0)
throw std::invalid_argument("Initial payment cannot be negative");
initialPayment_ = initialA;
}

inline double getInitialPayment() const { return initialPayment_; }

// Yearly interest rate
inline void setInterest(double i)
{
if (i <= 0)
throw std::invalid_argument("Interest must be > 0");
interest_ = i;
interestPeriodic_ = i / 100.0 / 12.0;
interestSet_ = true;
```

```
}

inline double getInterest() const { return interest_; }
inline double getPeriodicInterest() const { return interestPeriodic_; }

// Monthly payment
inline void setPayment(double P)
{
if (P <= 0)
throw std::invalid_argument("Payment must be > 0");
payment_ = P;
paymentSet_ = true;
}

inline double getPayment() const { return payment_; }

// Total loan period (N)
inline void setPeriodTotal(int N)
{
if (N <= 0)
throw std::invalid_argument("Total period must be > 0");
periodTotal_ = N;
#endif
```

## // File: LoanCalculator.cpp (Refactored)

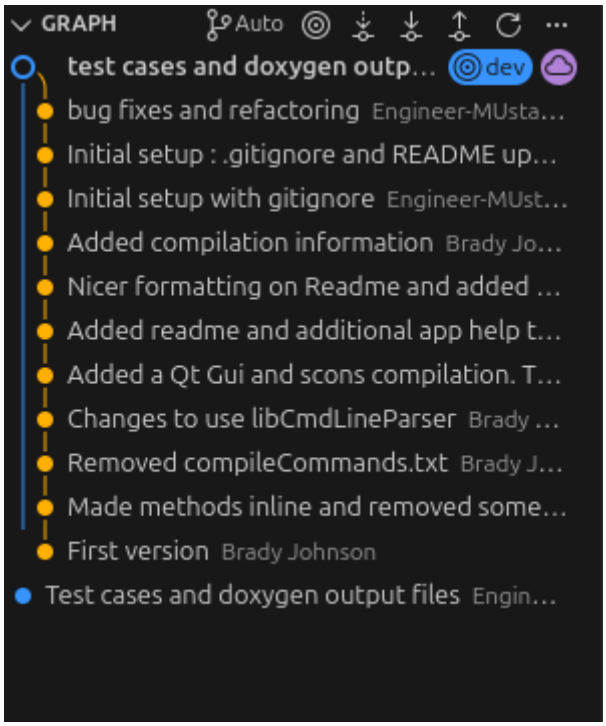
```
long double arg = 1.0L - (i * A / P);

if (!(arg > 0.0L))
throw std::invalid_argument("Payment too small, loan will never be repaid");

long double result = -log(arg) / log(1.0L + i);
return (double)result;
}

// Original Loan Amount
double LoanCalculator::calculateLoanAmount()
{
if (!paymentSet_ || !interestSet_ || !periodTotalSet_)
throw std::invalid_argument("Must set payment, interest and total period");

long double i = (long double)interestPeriodic_;

long double result = ((long double)payment_ / i) *
(1.0L - pow((1 + i), -periodTotal_));

return (double)result;
}

// Approximate Interest Rate
```

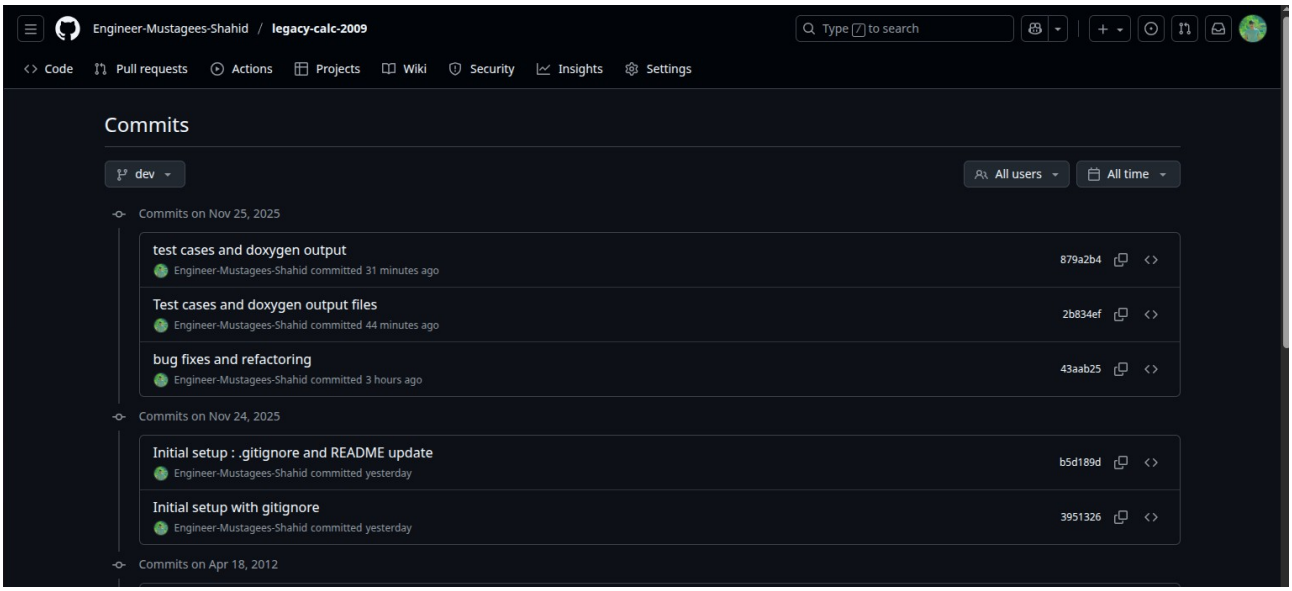
```
double LoanCalculator::calculateInterestRate()
{
if (!amountSet_ || !paymentSet_ || !periodTotalSet_)
throw std::invalid_argument("Must set amount, payment and total period");

long double q = log(1.0L + 1.0L / periodTotal_) / log(2.0L);
long double base = pow((1.0L + payment_ / amount_), 1.0L / q) - 1.0L;
long double monthlyInterest = pow(base, q) - 1.0L;

return (double)(monthlyInterest * 12 * 100);
}

// Effective Interest Rate
double LoanCalculator::calculateEffectiveInterestRate()
{
if (!amountSet_ || !periodTotalSet_)
throw std::invalid_argument("Must set amount and total period");

double payment = calculatePayment();
long double totalAmount = amount_ - initialPayment_;

long double q = log(1.0L + 1.0L / periodTotal_) / log(2.0L);
long double base = pow((1.0L + payment / totalAmount), 1.0L / q) - 1.0L;
long double monthlyInterest = pow(base, q) - 1.0L;

return (double)(monthlyInterest * 12 * 100);
}

std::string LoanCalculator::toString()
{
return "Loan Calculator (Refactored & Bug Fixed)";
}
```

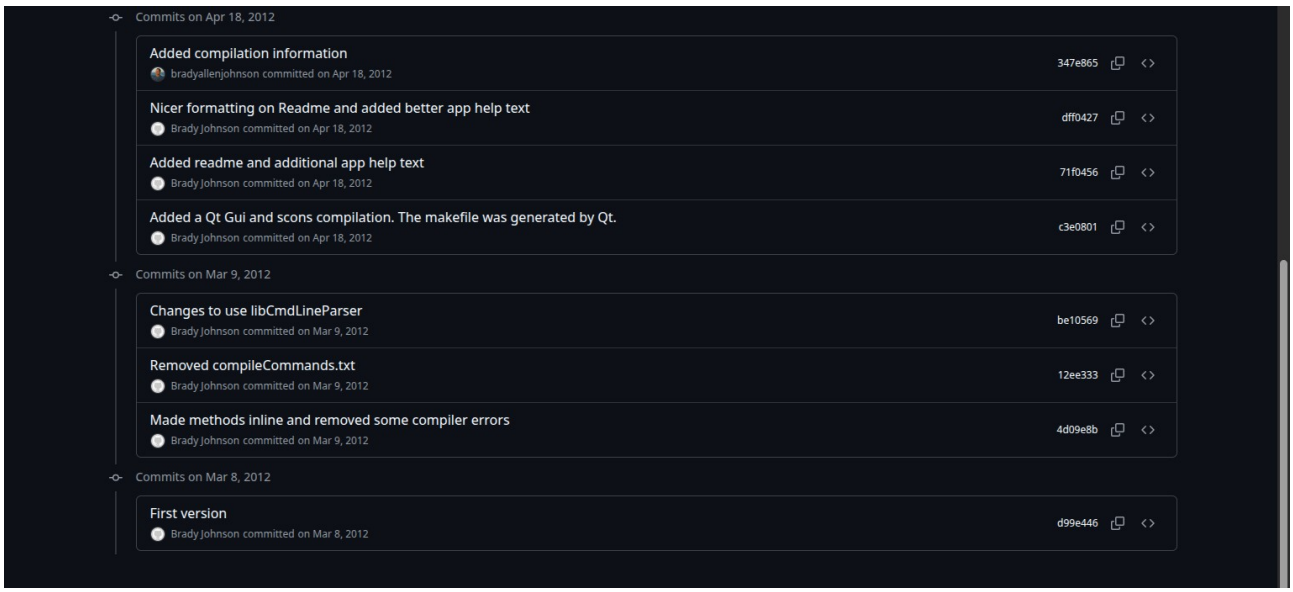
## Test Outputs :

```
mustagees-shahid@Hp-Elite-840:~/Documents/legacy-calc-2009-dev$ ./test
[==========] Running 3 tests from 1 test suite.
[----------] Global test environment set-up.
[----------] 3 tests from LoanCalculatorTest
[ RUN      ] LoanCalculatorTest.NormalEMICalculation
[       OK ] LoanCalculatorTest.NormalEMICalculation (0 ms)
[ RUN      ] LoanCalculatorTest.InvalidInputsThrow
[       OK ] LoanCalculatorTest.InvalidInputsThrow (0 ms)
[ RUN      ] LoanCalculatorTest.LargeTenureNoOverflow
[       OK ] LoanCalculatorTest.LargeTenureNoOverflow (0 ms)
[----------] 3 tests from LoanCalculatorTest (0 ms total)

[----------] Global test environment tear-down
[==========] 3 tests from 1 test suite ran. (0 ms total)
[  PASSED  ] 3 tests.
```

## Commits History :






**https://github.com/Engineer-Mustagees-Shahid/legacy-calc-2009/tree/dev**
**https://github.com/SaadAhmad99/legacy-calc-2009/tree/dev**